

⚡ Wiring the new ESP32-S3 board

For Yasir · Solar Monitor · board YD-ESP32-23 (ESP32-S3 N16R8) · 9 Oct 2026

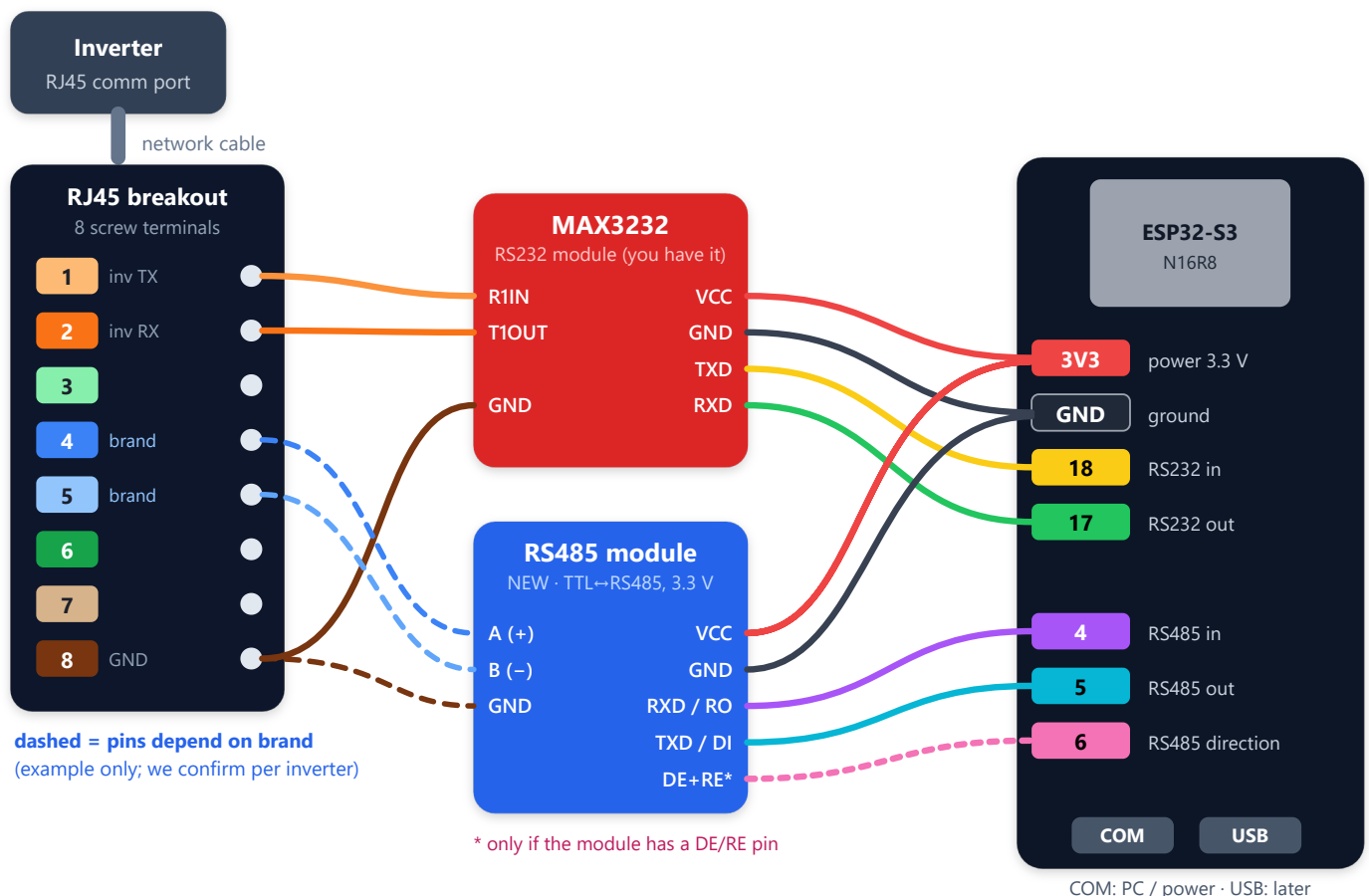
✓ Our inverters (Inverex Veyron, Galaxy): 3 wires is everything

Their RJ45 jack is an **RS232** port. RS232 has only 3 lines: **send, receive, ground**. The inverter sends **all** its data (solar, PV1/PV2, battery, grid, load, settings, faults) over those 3. Extra wires would carry nothing more.

+ Neighbours / other brands: add one small RS485 module now

Nitrox, Growatt, SolaX (Fronus), Solis, Sungrow, Huawei use a different "language" called **RS485 Modbus**, on **different RJ45 pins**, and every brand puts them on different pins. So we add a second module and a **screw-terminal RJ45 jack**, so any brand can be wired by moving 2–3 screw wires, without re-crimping.

1 The full picture (everything connected)



Inside the network cable the 8 colours are the standard ones (T568B: 1 orange-white, 2 orange, 3 green-white, 4 blue, 5 blue-white, 6 green, 7 brown-white, 8 brown). Always go by the **pin number**, not the colour: some cables are wired differently.

2 RS232 part: same as the C3 (do this today)

CABLE / RJ45	MAX3232
 Pin 1 inverter TX	R1IN (RS232 in)
 Pin 2 inverter RX	T1OUT (RS232 out)
 Pin 8 ground	GND (RS232 side)

MAX3232	ESP32-S3
 VCC	3V3
 GND	GND
 TXD	18 (was 10 on C3)
 RXD	17 (was 20 on C3)

If no data arrives after wiring, swap the two wires on **17** and **18** (some MAX3232 boards label TXD/RXD from the other side). Swapping these is safe.

3 RS485 part: for other brands (fit it now, use it later)

RS485 MODULE	ESP32-S3
 VCC	3V3
 GND	GND
 RXD / RO	4
 TXD / DI	5
 DE + RE (tied)	6 only if present

Cable side: A, B, GND

- A (+) and B (–) go to the two RS485 pins of **that brand's** port. GND to its ground pin, if it has one.
- Every brand is different (for example SolaX uses pins 4 and 5). We look it up in the brand's manual before connecting.
- If the readings are garbage, swap A and B. That is safe.
- Two boards share 3V3 and GND: use the board's two GND pins, or join the wires.

4 Why not just connect more wires to the same cable?

- **More wires ≠ more data.** The data comes from the **language** the inverter speaks. Voltronic (ours) speaks RS232 on pins 1, 2, 8 and sends everything there.
- Other brands speak **RS485**, on **other pins**, and sometimes on a **different port** (labelled RS485, COM, BMS, Meter or DRM). The RS485 module covers them.
- The **screw-terminal RJ45 breakout** lets us move wires per brand in 1 minute instead of cutting cables.
- Growatt SPF also has a **USB** port: that goes into the S3's **USB** socket later, no wires needed.

5 Safety rules

- **Never** plug our cable into a neighbour's inverter before we check its port. Send a photo of the **label** (brand + model) and the **ports** first.
- Never connect RS232 and RS485 to the **same** RJ45 pins at the same time. Only one module is wired to the cable per inverter.
- Never put the inverter cable straight onto the ESP pins (RS232 is ± 12 V). Always through a module.
- Modules run on **3V3**, never 5 V. Don't use pins 0, 3, 19, 20, 35, 36, 37, 43, 44, 45, 46, 48.
- Power the board from **one** USB cable only (COM port).

6 Shopping list (small extras)

PART	ASK FOR	APPROX.
RS485 module	"TTL to RS485 module, 3.3 V, auto direction" (or MAX3485 / SP3485 board)	Rs 150–250
RJ45 breakout	"RJ45 female to 8-pin screw terminal" ×2 (one spare)	Rs 100–200
Jumper wires	Female-to-female Dupont, 20 cm, mixed colours	Rs 100
Later (Growatt)	USB-C OTG to USB-A adapter + USB-A to USB-B cable	Rs 300

Today you can wire just section 2 (3 wires + 4 jumpers). That gives 100% of the data from our Inverex and Galaxy. Add section 3 whenever the parts arrive.